MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 3

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Jacobian Matrix

Let $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^m$ be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x})$$

$$\mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

Change of Variables Theorem (CoV)

Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^m$ be a C^1-diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

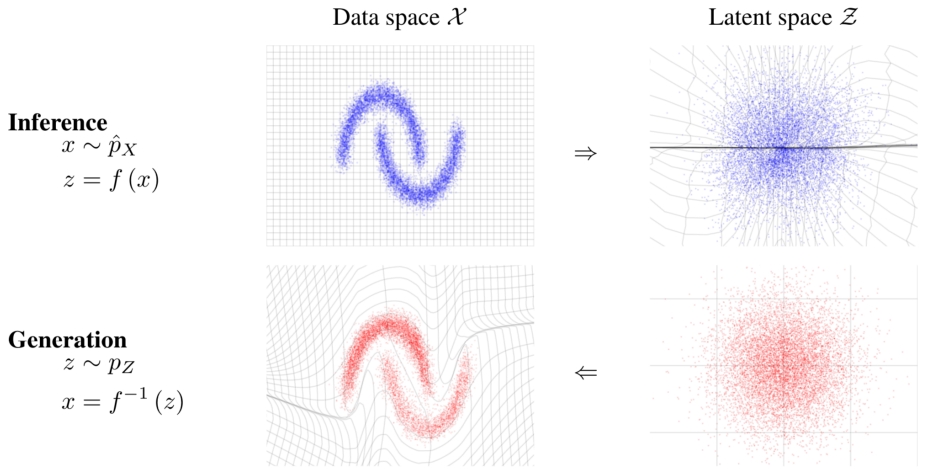
$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_\mathbf{f})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

$$p(\mathbf{z}) = p(\mathbf{x}) |\det(\mathbf{J}_{\mathbf{f}^{-1}})| = p(\mathbf{x}) \left| \det \left(\frac{\partial \mathbf{x}}{\partial \mathbf{z}} \right) \right| = p(\mathbf{f}^{-1}(\mathbf{z})) \left| \det \left(\frac{\partial \mathbf{f}^{-1}(\mathbf{z})}{\partial \mathbf{z}} \right) \right|$$

Recap of Previous Lecture

Definition

A normalizing flow is a C^1-diffeomorphism that transforms data $\mathbf{x}$ to noise $\mathbf{z}$.



Log-Likelihood

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \ldots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^{K} \log \left| \det(\mathbf{J}_{\mathbf{f}_k}) \right|$$

where $\mathbf{J}_{\mathbf{f}_k} = \dfrac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}}$.

Recap of Previous Lecture

NF Log-Likelihood

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x})) + \log \left| \det(\mathbf{J}_{\mathbf{f}}) \right|$$

The principal computational challenge is evaluating the Jacobian determinant.

$$\mathbf{z} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}$$

Structured Factors

Structured factorizations (e.g. LU or QR) simplify determinant and inverse calculations.

Keep triangular diagonals nonzero to preserve invertibility.

Learn the factors directly during training.

Kingma D. P., et al. Glow: Generative Flow with Invertible 1x1 Convolutions, 2018
Hoogeboom E., et al. Emerging Convolutions for Generative Normalizing Flows, 2019

Recap of Previous Lecture

Consider the autoregressive model:

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^{m} p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1}), \quad p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1}) = \mathcal{N}\left(\mu_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1}), \sigma^2_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1})\right)$$

Gaussian AR NF

$$\mathbf{x} = \mathbf{f}_{\boldsymbol{\theta}}^{-1}(\mathbf{z}) \quad \Rightarrow \quad x_j = \sigma_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1}) \cdot z_j + \mu_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1}).$$

$$\mathbf{z} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}) \quad \Rightarrow \quad z_j = (x_j - \mu_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1})}.$$

- This gives a C^1-**diffeomorphism** from $p(\mathbf{z})$ to $p_{\boldsymbol{\theta}}(\mathbf{x})$ (assume that $\sigma_j \neq 0$).
- The Jacobian matrix of this transformation is triangular.

Sampling must be done sequentially, but density evaluation can be parallelized.

Recap of Previous Lecture

Split $\mathbf{x}$ and $\mathbf{z}$ into two parts:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}]$$

Coupling Layer

$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1 \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \boldsymbol{\sigma}_{\boldsymbol{\theta}}(\mathbf{z}_1) + \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z}_1) \end{cases} \qquad \begin{cases} \mathbf{z}_1 = \mathbf{x}_1 \\ \mathbf{z}_2 = (\mathbf{x}_2 - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_1)) \odot \frac{1}{\boldsymbol{\sigma}_{\boldsymbol{\theta}}(\mathbf{x}_1)} \end{cases}$$

In both training and sampling, only a single forward pass is needed!

Jacobian

$$\det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & 0_{d \times m-d} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix} = \prod_{j=1}^{m-d} \frac{1}{\sigma_{j,\boldsymbol{\theta}}(\mathbf{x}_1)}.$$

A coupling layer is a special instance of the Gaussian autoregressive NF.

TarFlow uses affine autoregressive transforms on image patches. A causal Transformer predicts componentwise shifts and positive scales from earlier patches. The Jacobian remains triangular; sampling is sequential across patches and parallel within each patch.

Dinh L., Sohl-Dickstein J., Bengio S. Density Estimation Using Real NVP, 2016
Zhai S. et al. Normalizing Flows are Capable Generative Models, 2024

Recap of Previous Lecture

A flow evaluates a **continuous density**; an 8-bit image has discrete pixel values.

Probability of a Pixel Bin

For $\mathbf{x} \in \{0, \dots, 255\}^m$ and a model supported on $[0, 256)^m$:

$$P_{\boldsymbol{\theta}}(\mathbf{x}) = \int_{[0,1)^m} p_{\boldsymbol{\theta}}(\mathbf{x} + \mathbf{u})\, d\mathbf{u}.$$

Uniform Dequantization

$$\mathbf{u} \sim \mathrm{Uniform}([0, 1)^m), \qquad \mathbf{v} = \mathbf{x} + \mathbf{u}, \qquad \lfloor \mathbf{v} \rfloor = \mathbf{x}.$$

Train the flow on $\mathbf{v}$. Jensen's inequality gives

$$\log P_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{\mathbf{u}} \left[\log p_{\boldsymbol{\theta}}(\mathbf{x} + \mathbf{u}) \right].$$

Exact continuous density evaluation gives a **lower bound** on discrete log-likelihood.

Theis L., van den Oord A., Bethge M. A Note on the Evaluation of Generative Models, 2015. Section 3.1.

Recap of Previous Lecture

Maximum Likelihood Estimation (MLE) Problem

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} p_{\boldsymbol{\theta}}(\mathbf{X}) = \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{n} p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i).$$

The distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ should be highly complex (just like the true data distribution $p_{\text{data}}(\mathbf{x})$).

Latent Variable

Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \int p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z}) d\mathbf{z}.$$

- $p(\mathbf{z})$ is prior distribution over the latent variable.
- $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ is **decoder** distribution.
- Both $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ are usually much simpler than $p_{\boldsymbol{\theta}}(\mathbf{x})$.

Outline

01 Latent Variable Models (LVM) (continued)

02 Variational Evidence Lower Bound (ELBO)

03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

Outline

01 **Latent Variable Models (LVM) (continued)**

02 Variational Evidence Lower Bound (ELBO)

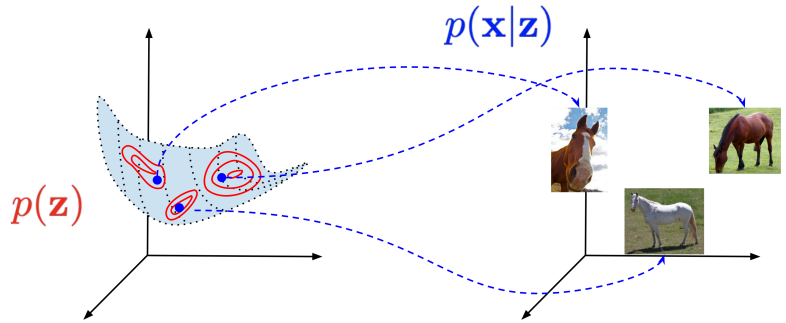
03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

MLE for LVM

$$\sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \sum_{i=1}^{n} \log \int p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \to \max_{\boldsymbol{\theta}}.$$



A Naive Lower Bound

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^{K} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_k),$$

where $\mathbf{z}_k \sim p(\mathbf{z})$.

- **Observation-independent samples:** the prior does not adapt to $\mathbf{x}$.
- **Potentially loose bound:** more samples reduce noise, but do not tighten the bound.

Image credit: https://jmtomczak.github.io/blog/4/4_VAE.html

Outline

01 Latent Variable Models (LVM) (continued)

02 Variational Evidence Lower Bound (ELBO)

03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

ELBO Derivation I

Bayes theorem

$$p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x}) = \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{x})} \quad \Rightarrow \quad p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}$$

Introduce an auxiliary distribution $q(\mathbf{z})$ to build a more flexible family of lower bounds.

$$\int q(\mathbf{z}) d\mathbf{z} = 1, \qquad \operatorname{supp}(q(\mathbf{z})) = \operatorname{supp}(p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) = \mathbb{R}^d.$$

Split the logarithm

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log \left[\frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \cdot \frac{q(\mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})} \right] = \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} + \log \frac{q(\mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}.$$

Average over $\mathbf{z} \sim q$

Since $\log p_{\boldsymbol{\theta}}(\mathbf{x})$ does not depend on $\mathbf{z}$:

$$\boxed{\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} + \mathbb{E}_q \log \frac{q(\mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}}.$$

ELBO Derivation II

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} + \mathbb{E}_q \log \frac{q(\mathbf{z})}{p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})}$$

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + \underbrace{D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x}))}_{\geq 0}$$

The KL term is the **exact gap** between log-likelihood and ELBO.

Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

The bound holds for any admissible $q(\mathbf{z})$.

Maximize the ELBO as a surrogate for log-likelihood:

$$\max_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) \quad \longrightarrow \quad \max_{q,\boldsymbol{\theta}} \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x})$$

For fixed $\boldsymbol{\theta}$, the **variational parameter** q controls the tightness of the bound.

Variational Evidence Lower Bound (ELBO)

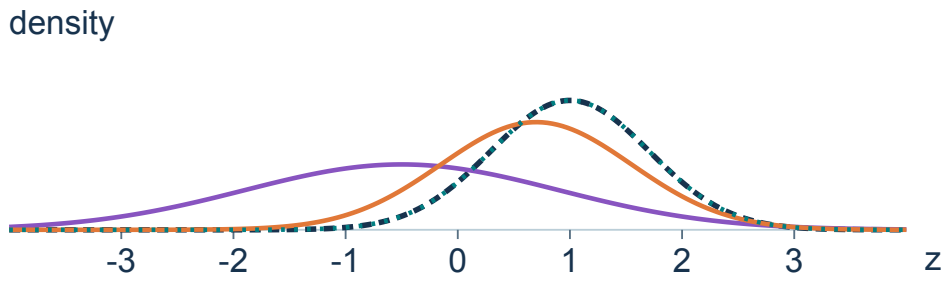
$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x}))$$

What is the optimal distribution $q^*(\mathbf{z})$ given fixed $\boldsymbol{\theta}^*$?

$$q^*(\mathbf{z}) = \arg\max_q \mathcal{L}_{q,\boldsymbol{\theta}^*}(\mathbf{x}) = \arg\min_q D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})) = p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x}).$$

Here we got the intuition about variational distribution $q(\mathbf{z})$: it estimates the posterior $p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})$.

$p(z|x) = \mathcal{N}(1, 1/2)$



$p(z) = \mathcal{N}(0,1), \quad p(x|z) = \mathcal{N}(z,1), \quad x = 2$

Poor	Better	Exact
$\mu_q = -0.5, \quad \sigma_q = 1.400$	$\mu_q = 0.7, \quad \sigma_q = 0.850$	$\mu_q = 1, \quad \sigma_q = 0.707$
ELBO -5.292	ELBO -2.394	ELBO -2.266
KL 3.027	KL 0.128	KL 0.000

log p(x) = -2.266 stays fixed. The exact posterior closes the gap.

Variational Evidence Lower Bound (ELBO)

Factor the joint distribution to separate reconstruction from the prior penalty.

$$\begin{aligned}\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) &= \int q(\mathbf{z}) \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \int q(\mathbf{z}) \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) d\mathbf{z} + \int q(\mathbf{z}) \log \frac{p(\mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z})\|p(\mathbf{z}))\end{aligned}$$

Log-Likelihood Decomposition

Substitute this form into the earlier identity. The KL to the posterior remains the ELBO gap.

$$\begin{aligned}\log p_{\boldsymbol{\theta}}(\mathbf{x}) &= \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q(\mathbf{z})\|p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \\ &= \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z})\|p(\mathbf{z})) + D_{\mathrm{KL}}(q(\mathbf{z})\|p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})).\end{aligned}$$

- What do we get if we drop $D_{\mathrm{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$ term from ELBO?
- What do we have in the term $\mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ in the case of Normal distribution?

ELBO = reconstruction − regularization: explain $\mathbf{x}$ well; penalize deviations of q from the prior.

ELBO: Key Takeaways

A lower bound for any admissible q

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

An exact gap

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) - \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \geq 0$$

For fixed $\boldsymbol{\theta}$, the bound is tight when $q(\mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$.

Reconstruction − regularization

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \underbrace{\mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})}_{\text{Reconstruction}} - \underbrace{D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z}))}_{\text{Regularization}}$$

Joint optimization

$$\max_{\boldsymbol{\theta}, q} \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x})$$

Outline

01 Latent Variable Models (LVM) (continued)

02 Variational Evidence Lower Bound (ELBO)

03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

Parametric Variable Posterior

Variational Posterior

$$q^*(\mathbf{z}) = \arg\max_q \mathcal{L}_{q,\boldsymbol{\theta}^*}(\mathbf{x}) = \arg\min_q D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})) = p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x}).$$

- $p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})$ may be **intractable**;
- $q(\mathbf{z})$ is individual for each data point $\mathbf{x}$.

Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, conditioned on data $\mathbf{x}$ and parameterized by $\boldsymbol{\phi}$.

Gradient Update

$$\begin{bmatrix} \boldsymbol{\phi}_k \\ \boldsymbol{\theta}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{\phi}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \\ \boldsymbol{\theta}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \end{bmatrix} \Bigg|_{(\boldsymbol{\phi}_{k-1}, \boldsymbol{\theta}_{k-1})}$$

Amortized Inference: One Shared Encoder

Optimize separately for each observation

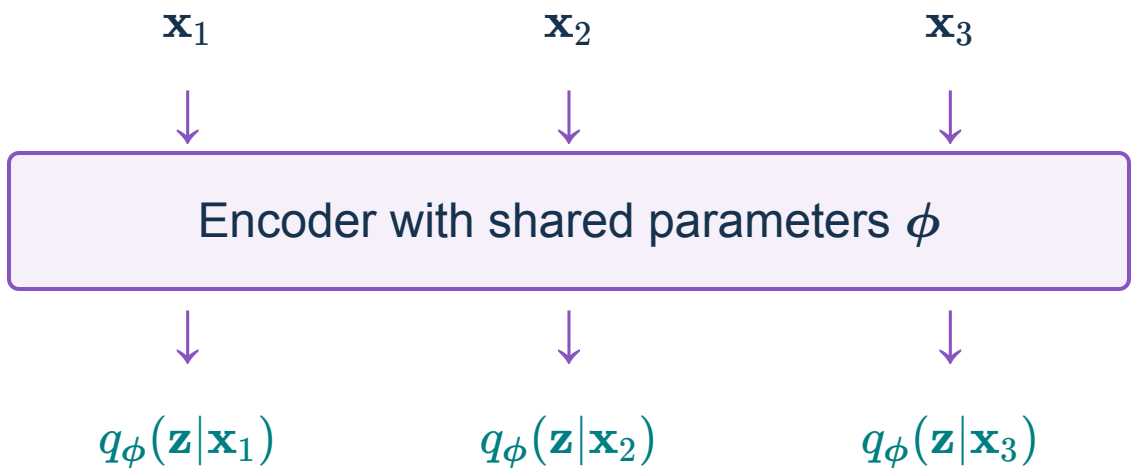
$\mathbf{x}_1 \rightarrow$ Fit parameters of q_1 $\rightarrow q_1(\mathbf{z})$

$\mathbf{x}_2 \rightarrow$ Fit parameters of q_2 $\rightarrow q_2(\mathbf{z})$

$\mathbf{x}_3 \rightarrow$ Fit parameters of q_3 $\rightarrow q_3(\mathbf{z})$

A separate optimization problem for each new input.

Learn one shared mapping



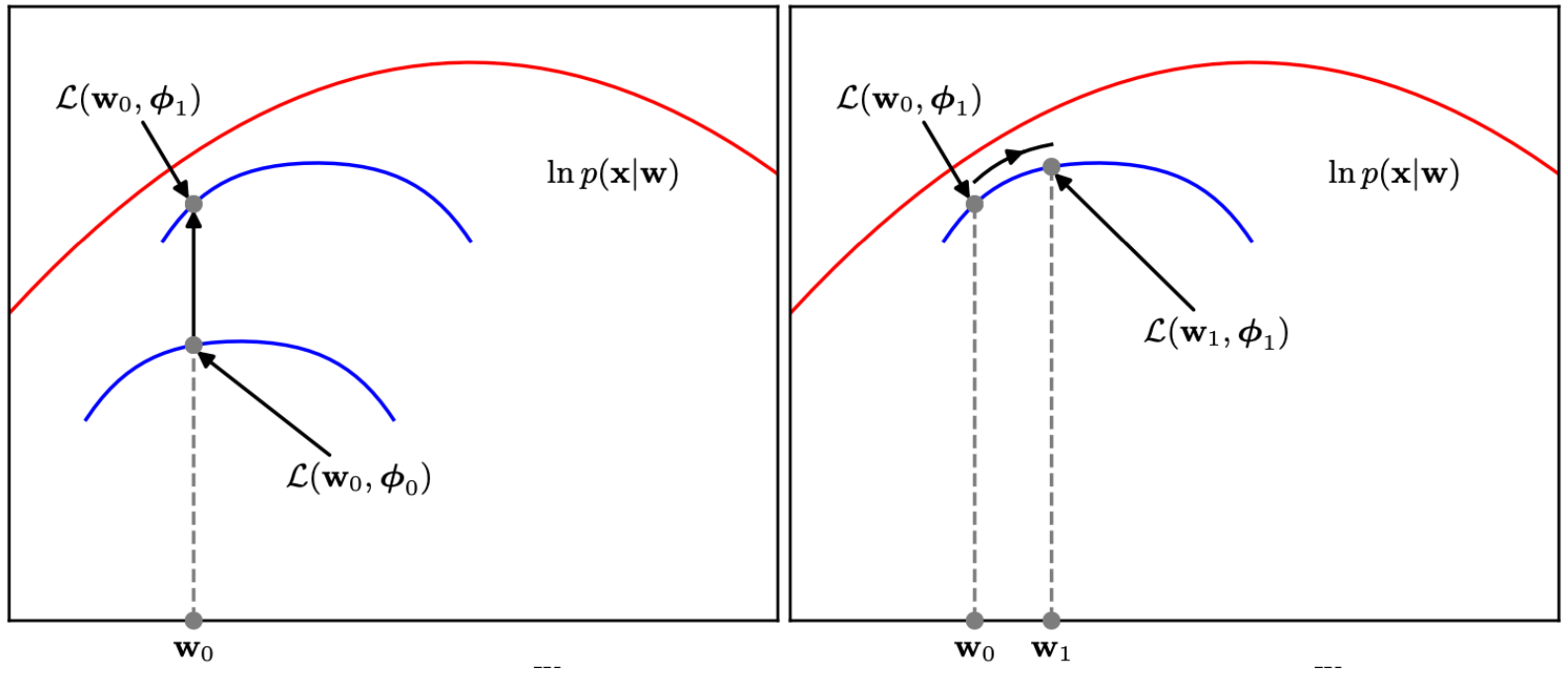
After training, a forward pass predicts the variational distribution for a new input.

> The variational distribution is conditioned on each input. The same parameters ϕ are reused to produce all of them.

ELBO Optimization

Gradient Update

$$\begin{bmatrix}\boldsymbol{\phi}_k \\ \boldsymbol{\theta}_k\end{bmatrix} = \begin{bmatrix}\boldsymbol{\phi}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \\ \boldsymbol{\theta}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})\end{bmatrix}\Bigg|_{(\boldsymbol{\phi}_{k-1},\boldsymbol{\theta}_{k-1})}$$



For fixed $\boldsymbol{\theta}$, optimizing $\boldsymbol{\phi}$ within a restricted family need not close the KL gap.

ELBO Optimization

ELBO

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}).$$

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

Gradient Update

$$\begin{bmatrix}\boldsymbol{\phi}_k \\ \boldsymbol{\theta}_k\end{bmatrix} = \left.\begin{bmatrix}\boldsymbol{\phi}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \\ \boldsymbol{\theta}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})\end{bmatrix}\right|_{(\boldsymbol{\phi}_{k-1},\boldsymbol{\theta}_{k-1})}$$

- $\boldsymbol{\phi}$ denotes the parameters of the variational posterior $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ (encoder).
- $\boldsymbol{\theta}$ represents the parameters of the generative model $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ (decoder).

The remaining step is to obtain **unbiased** Monte Carlo estimates of the gradients: $\nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$ and $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$.

Outline

01 Latent Variable Models (LVM) (continued)

02 Variational Evidence Lower Bound (ELBO)

03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

ELBO Gradients: $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

Gradient $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x})$

$q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ is independent of $\boldsymbol{\theta}$. Differentiate the decoder.

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) &= \nabla_{\boldsymbol{\theta}} \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\
&= \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\
&\approx \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}).
\end{aligned}$$

Naive Monte Carlo Estimation

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \int \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \approx \frac{1}{K} \sum_{k=1}^{K} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_k), \quad \mathbf{z}_k \sim p(\mathbf{z}).$$

Prior samples do not depend on $\mathbf{x}$ and may reconstruct it poorly. We train $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ to favor latent values that reconstruct this particular observation.

Image credit: https://jmtomczak.github.io/blog/4/4_VAE.html

ELBO Gradients: $\nabla_{\phi}\mathcal{L}_{\phi,\theta}(\mathbf{x})$

Gradient $\nabla_{\phi}\mathcal{L}_{\phi,\theta}(\mathbf{x})$

Unlike the $\boldsymbol{\theta}$-gradient, we must account for **the dependence of $q_{\phi}(\mathbf{z}|\mathbf{x})$ on ϕ**:

$$\nabla_{\phi}\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \nabla_{\phi}\int q_{\phi}(\mathbf{z}|\mathbf{x})\log p_{\theta}(\mathbf{x}|\mathbf{z})d\mathbf{z} - \nabla_{\phi}D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

$$\neq \int q_{\phi}(\mathbf{z}|\mathbf{x})\nabla_{\phi}\log p_{\theta}(\mathbf{x}|\mathbf{z})d\mathbf{z} - \nabla_{\phi}D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

Reparametrization: A Gaussian Example

Let the **encoder** predict the mean and standard deviation of a Gaussian:

$$q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}^2_{\phi}(\mathbf{x})).$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \qquad \mathbf{z} = \boldsymbol{\mu}_{\phi}(\mathbf{x}) + \boldsymbol{\sigma}_{\phi}(\mathbf{x}) \odot \boldsymbol{\epsilon}.$$

Sample standard Gaussian noise, then **scale and shift it**.

Average the reconstruction term over noise whose distribution **does not depend on ϕ**:

$$\mathbb{E}_{\mathbf{z}\sim q_{\phi}(\mathbf{z}|\mathbf{x})}\log p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\log p_{\theta}\left(\mathbf{x}|\boldsymbol{\mu}_{\phi}(\mathbf{x}) + \boldsymbol{\sigma}_{\phi}(\mathbf{x}) \odot \boldsymbol{\epsilon}\right).$$

Reparametrization Trick

From the Gaussian Example to a General Transformation

Write the previous scale and shift as $\mathbf{z} = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})$. More generally, let a **differentiable** $\mathbf{g}_{\boldsymbol{\phi}}$ transform noise $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$ into samples $\mathbf{z} \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, with $p(\boldsymbol{\epsilon})$ independent of $\boldsymbol{\phi}$.

The expectation is unchanged (LOTUS):

$$\mathbb{E}_{\mathbf{z} \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \mathbf{f}(\mathbf{z}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})} \mathbf{f}(\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})).$$

Differentiate through the Transformation

$$\begin{aligned} \nabla_{\boldsymbol{\phi}} \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \mathbf{f}(\mathbf{z}) d\mathbf{z} &= \nabla_{\boldsymbol{\phi}} \int p(\boldsymbol{\epsilon}) \mathbf{f}(\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})) d\boldsymbol{\epsilon} \\ &= \int p(\boldsymbol{\epsilon}) \nabla_{\boldsymbol{\phi}} \mathbf{f}(\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})) d\boldsymbol{\epsilon} \approx \nabla_{\boldsymbol{\phi}} \mathbf{f}(\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon}^*)), \end{aligned}$$

where $\boldsymbol{\epsilon}^* \sim p(\boldsymbol{\epsilon})$.

> During backpropagation, **keep $\boldsymbol{\epsilon}^*$ fixed** and differentiate through $\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon}^*)$.

ELBO Gradients: $\nabla_{\phi}\mathcal{L}_{\phi,\theta}(\mathbf{x})$

$$\nabla_{\phi}\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \nabla_{\phi}\int q_{\phi}(\mathbf{z}|\mathbf{x})\log p_{\theta}(\mathbf{x}|\mathbf{z})d\mathbf{z} - \nabla_{\phi}D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

Reconstruction Term (Reparametrization Trick)

$$\nabla_{\phi}\int q_{\phi}(\mathbf{z}|\mathbf{x})\log p_{\theta}(\mathbf{x}|\mathbf{z})d\mathbf{z} = \int p(\boldsymbol{\epsilon})\nabla_{\phi}\log p_{\theta}(\mathbf{x}|\mathbf{g}_{\phi}(\mathbf{x},\boldsymbol{\epsilon}))d\boldsymbol{\epsilon}$$

$$\approx \nabla_{\phi}\log p_{\theta}\left(\mathbf{x}|\boldsymbol{\sigma}_{\phi}(\mathbf{x})\odot\boldsymbol{\epsilon}^{*} + \boldsymbol{\mu}_{\phi}(\mathbf{x})\right),$$

where $\boldsymbol{\epsilon}^{*}\sim\mathcal{N}(0,\mathbf{I})$; **differentiate through** $\mathbf{g}_{\phi}(\mathbf{x},\boldsymbol{\epsilon}^{*})$.

The **encoder** predicts $\boldsymbol{\mu}_{\phi}(\mathbf{x})$ and $\boldsymbol{\sigma}_{\phi}(\mathbf{x})$; the **decoder** defines $p_{\theta}(\mathbf{x}|\mathbf{z})$.
To update ϕ, backpropagate $\log p_{\theta}(\mathbf{x}|\mathbf{z})$ through $\mathbf{z}$, keeping θ fixed.

KL Term

$p(\mathbf{z})$ is the prior over latents $\mathbf{z}$, typically $p(\mathbf{z})=\mathcal{N}(0,\mathbf{I})$.

$$\nabla_{\phi}D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) = \nabla_{\phi}D_{\mathrm{KL}}\left(\mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}),\boldsymbol{\sigma}^{2}_{\phi}(\mathbf{x}))\|\mathcal{N}(0,\mathbf{I})\right)$$

This expression admits a closed-form analytic solution.

Outline

01 Latent Variable Models (LVM) (continued)

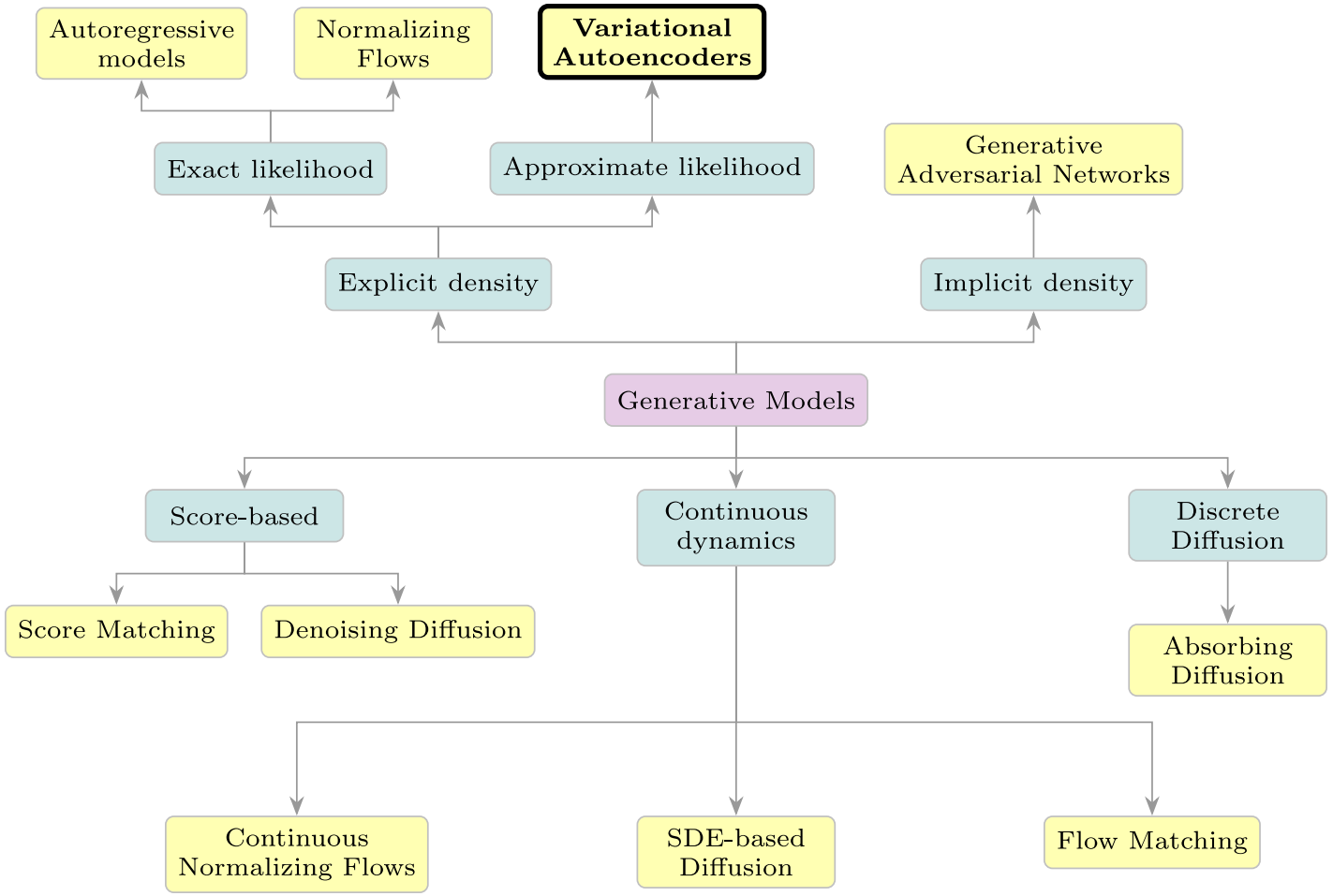
02 Variational Evidence Lower Bound (ELBO)

03 Amortized Inference

04 ELBO Gradients, Reparametrization Trick

05 Variational Autoencoder (VAE)

Generative Models Taxonomy



Variational Autoencoder (VAE)

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$.
2. Reparametrize $\mathbf{z} = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})$.
3. Compute the ELBO:

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \approx \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})).$$

4. Update $\boldsymbol{\phi}$, $\boldsymbol{\theta}$ via stochastic gradient ascent.

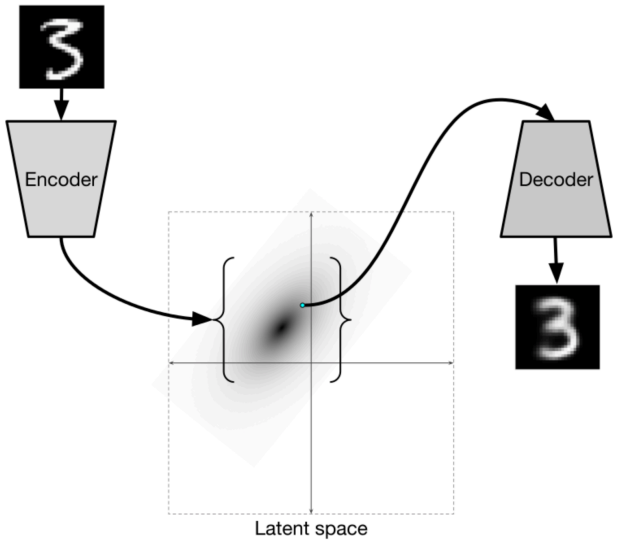
Sampling

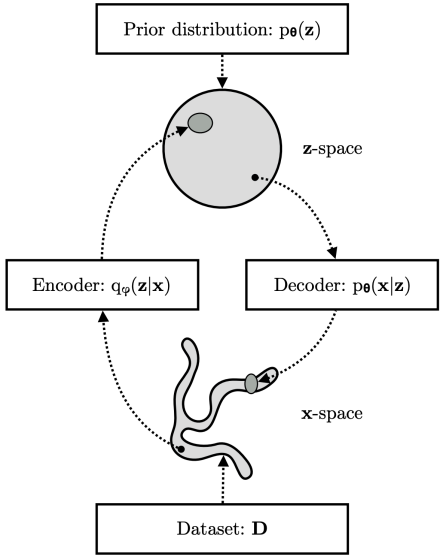
1. Sample $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.
2. Sample $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$.

Note: The encoder $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Variational Autoencoder

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$



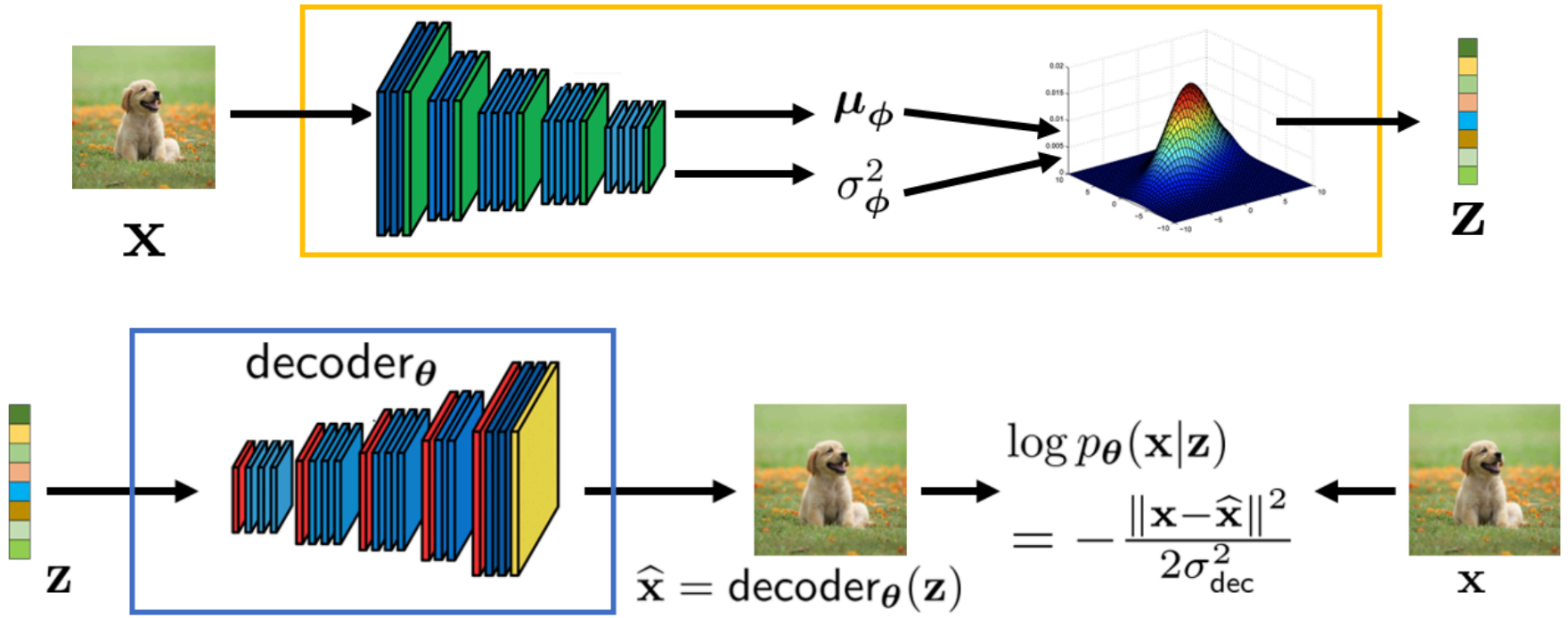


VAEs are widely used as a preliminary stage of projecting data onto low-dimensional space.

image credit: http://ijdykeman.github.io/ml/2016/12/21/cvae.html
Kingma D. P., Welling M., An Introduction to Variational Autoencoders, 2019

Variational Autoencoder

- The **encoder** predicts $\boldsymbol{\mu}_\phi(\mathbf{x})$ and $\boldsymbol{\sigma}_\phi(\mathbf{x})$, which parameterize $q_\phi(\mathbf{z}|\mathbf{x})$.
- The **decoder** predicts the parameters of the observed data distribution $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$.



VAE vs Normalizing Flows

	VAE	NF
Objective	ELBO $\mathcal{L}$	Forward KL/MLE
Encoder	stochastic $\mathbf{z} \sim q_{\phi}(\mathbf{z}\|\mathbf{x})$	deterministic $\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x})$ $q_{\theta}(\mathbf{z}\|\mathbf{x}) = \delta(\mathbf{z} - \mathbf{f}_{\theta}(\mathbf{x}))$
Decoder	stochastic $\mathbf{x} \sim p_{\theta}(\mathbf{x}\|\mathbf{z})$	deterministic $\mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z})$ $p_{\theta}(\mathbf{x}\|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{f}_{\theta}^{-1}(\mathbf{z}))$
Parameters	ϕ, θ	$\theta \equiv \phi$

Theorem

MLE for a normalizing flow is equivalent to maximizing the ELBO for a VAE where:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{f}_{\theta}^{-1}(\mathbf{z})); \quad q_{\theta}(\mathbf{z}|\mathbf{x}) = \delta(\mathbf{z} - \mathbf{f}_{\theta}(\mathbf{x})).$$

Nielsen D., et al., SurVAE Flows: Surjections to Bridge the Gap Between VAEs and Flows, 2020

Summary

- LVMs maximize the ELBO as a surrogate for log-likelihood; a restricted variational family can leave a nonzero KL gap.
- Parametric posterior distribution $q_{\phi}(\mathbf{z}|\mathbf{x})$ makes the method scalable.
- The reparametrization trick provides unbiased gradients with respect to the variational posterior $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- A VAE combines a stochastic encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ with a stochastic decoder $p_{\theta}(\mathbf{x}|\mathbf{z})$.
- Sampling uses the prior and decoder; the encoder is not needed.